SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43
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REGISTER NUMBER: 192324244

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. (L2)

Assessment Tool : Case Study

Weightage: 10%

QUESTIONS:

Scenario: Web Development Standards Implementation

A startup is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

```
A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>
```

Snippet 2: Form Submission Methods

```
A. GET
B. POST
C. PUT
D. DELETE
```

Snippet 3: Navigation Elements

```
A. <div>
B. <nav>
C. <section>
D. <header>
```

Snippet 4: XML Syntax

```
A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note
```

Snippet 5: Web Communication Protocols

```
A. FTP
B. HTTP
C. SMTP
D. SNMP
```

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer.
2. Identify the **appropriate HTTP method for form submission** and explain why.
3. Choose the **proper semantic element for navigation** and justify its usage.
4. Identify the **correct XML syntax** and explain the error in other options.
5. Choose the **correct protocol for web communication** and justify your choice.

SOLUTION:

1. Select the correct HTML tag for video embedding

Answer: Option B – `<video src="video.mp4" controls></video>`

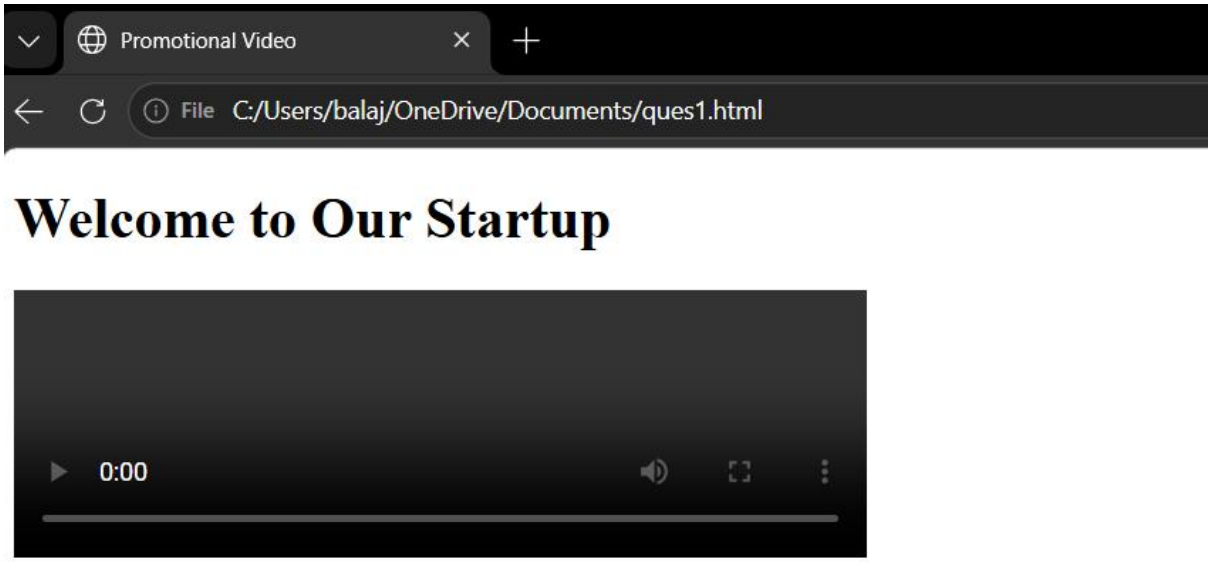
The `<video>` element is the standard HTML5 way to embed a video directly in a web page without any external plugin. It is the only valid tag among the given options, so the promotional video can be played in all modern browsers.

- `src` – specifies the location of the video file (video.mp4).
- `controls` – displays the browser's built-in play/pause, volume, seek bar and full-screen buttons.
- Text placed between the opening and closing tags acts as fallback content for browsers that do not support video.
- Options A, C and D (`<media>`, `<movie>`, `<vid>`) are not defined in the HTML5 specification. Browsers ignore unknown elements, so no video would be displayed.

HTML CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Promotional Video</title>
</head>
<body>
  <h1>Welcome to Our Startup</h1>
  <video src="video.mp4" controls width="480">
    Your browser does not support the video tag.
  </video>
</body>
</html>
```

OUTPUT:



2. Identify the appropriate HTTP method for form submission

Answer: Option B – POST

A registration form sends personal data such as a name, an email address and a password. The POST method is the appropriate choice because it places the form data in the body of the HTTP request instead of the URL.

- **Data stays out of the URL:** with POST the values are not shown in the address bar, browser history, bookmarks or server access logs.
- **Suitable for sending data:** POST is meant for submitting data that creates or changes something on the server, such as a new user account, and it has no practical limit on data size.
- **GET** appends the data to the URL as a query string. It is intended for retrieving information, so it is unsuitable for passwords and registration data.
- **PUT** and **DELETE** are used to replace and remove resources. HTML forms support only GET and POST, so these two cannot be used in the `method` attribute of a form.

- POST alone does not encrypt the data. To send it **securely**, the site must also use HTTPS.

HTML CODE:

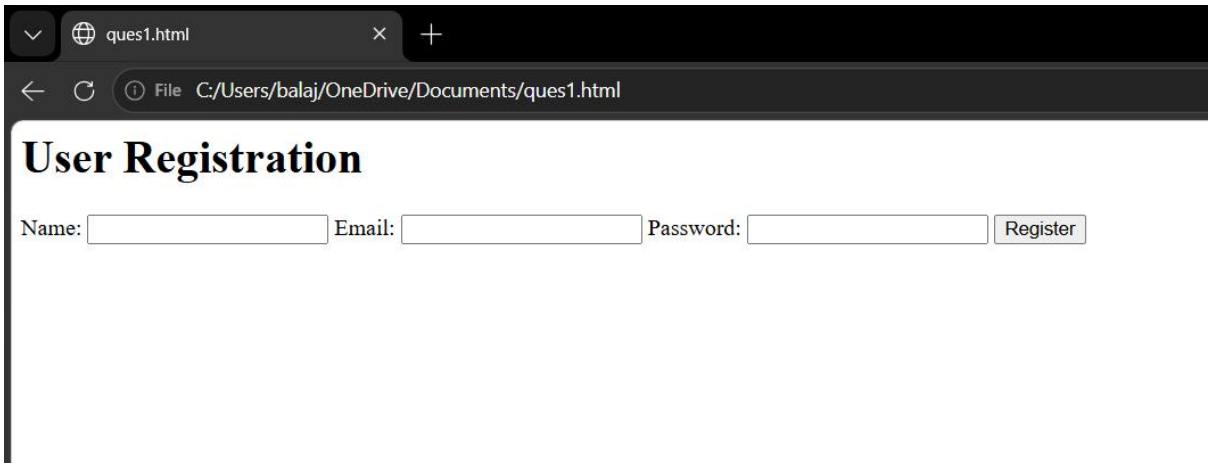
```
<form action="/register" method="POST">
  <label for="name">Name:</label>
  <input type="text" id="name" name="name" required>

  <label for="email">Email:</label>
  <input type="email" id="email" name="email" required>

  <label for="pwd">Password:</label>
  <input type="password" id="pwd" name="password" required>

  <button type="submit">Register</button>
</form>
```

OUTPUT:



3. Choose the proper semantic element for navigation

Answer: Option B – `<nav>`

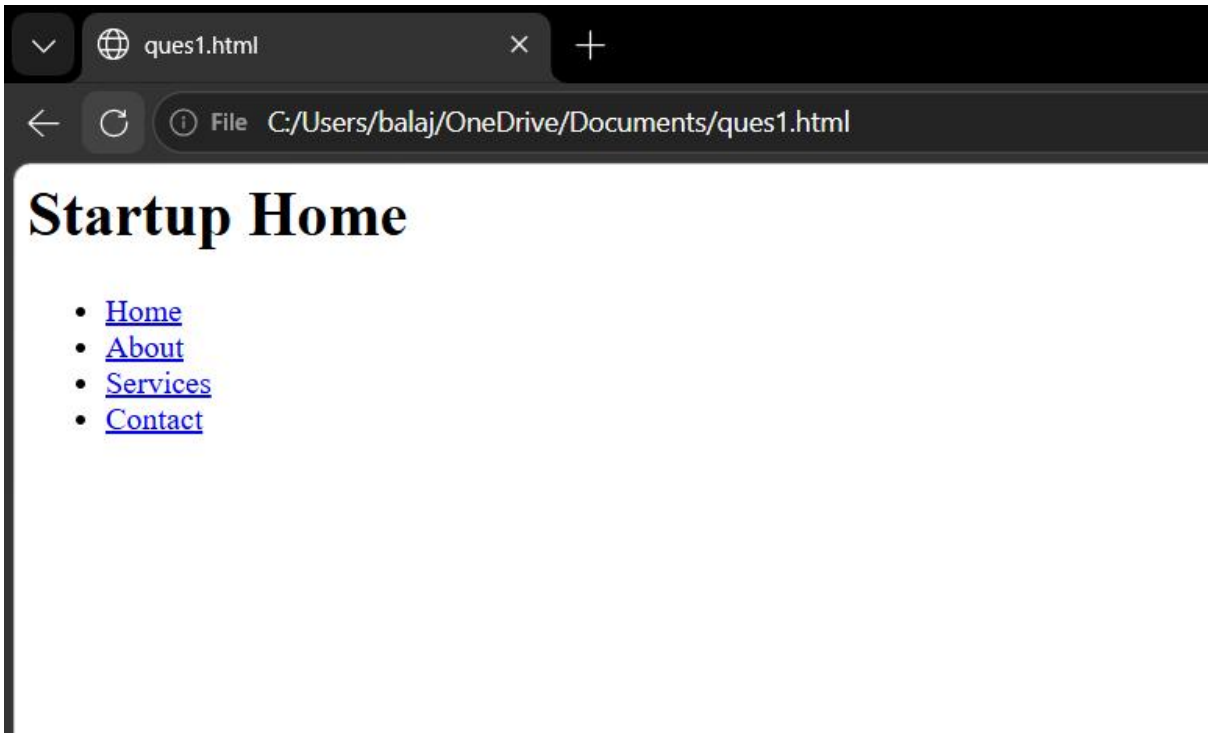
The `<nav>` element is the HTML5 semantic element for a block of major navigation links, such as the main menu of a website. It tells browsers, search engines and assistive technologies that the enclosed links are used to move around the site.

- **Accessibility:** screen readers announce `<nav>` as a navigation landmark, so users can jump directly to the menu or skip it.
- **Meaningful structure:** the markup describes its purpose, which makes the code easier to read and maintain and helps search engines understand the site structure.
- `<div>` is a generic container with no meaning, so it does not follow semantic standards.
- `<section>` groups related content under a heading; it does not indicate navigation.
- `<header>` holds introductory content such as a logo or title. It may contain a `<nav>`, but it is not itself a navigation element.

HTML CODE:

```
<header>
  <h1>Startup Home</h1>
</header>
<nav>
  <ul>
    <li><a href="index.html">Home</a></li>
    <li><a href="about.html">About</a></li>
    <li><a href="services.html">Services</a></li>
    <li><a href="contact.html">Contact</a></li>
  </ul>
</nav>
```

OUTPUT:



4. Identify the correct XML syntax and explain the error in other options

Answer: Option B – `<note><to>User</to><from>Admin</from></note>`

Option B is a well-formed XML document. It has a single root element (`<note>`), every start tag has a matching end tag, and the elements are properly nested. The other options break these rules:

- **Option A:** the root element `<note>` is never closed because the `</note>` end tag is missing.
- **Option C:** the tags are mismatched. `<to>` is closed with `</from>`, and `<from>` was never opened. XML requires each end tag to match its start tag exactly.
- **Option D:** the end tag is incomplete (`</note` has no closing `>`), so the element is not properly terminated.

Basic rules of well-formed XML: one root element, every element closed, correct nesting, case-sensitive tag names, and quoted attribute values.

XML CODE:

```
<?xml version="1.0" encoding="UTF-8"?>
<note>
  <to>User</to>
  <from>Admin</from>
</note>
```

OUTPUT:

This XML file does not appear to have any style information associated with it. The document tree is shown below.

```
▼<note>
   <to>User</to>
   <from>Admin</from>
 </note>
```

5. Choose the correct protocol for web communication

Answer: Option B – HTTP (Hypertext Transfer Protocol)

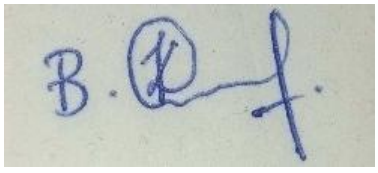
HTTP is the application-layer protocol used by web browsers and web servers to exchange web content such as HTML pages, images, videos and form data. The browser sends an HTTP request and the server returns an HTTP response with a status code and the requested content.

- **FTP** is used for transferring files between computers, not for delivering web pages to a browser.
- **SMTP** is used for sending email between mail servers.
- **SNMP** is used for monitoring and managing network devices such as routers and switches.
- Because the startup's application handles registration data, it should use **HTTPS** (HTTP over TLS/SSL), which encrypts the communication between the browser and the server.

Thus, using the video element, POST method, nav element, well-formed XML and HTTP together helps the startup build a standards-compliant and reliable web application.

Code of Conduct:

I **B KIRTHANAA 192324244** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student: B KIRTHANAA

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand